

When Ambitions Align: Peer Parental Aspiration and Student Achievement¹

Yuzhi Zhou²

Abstract

Parental aspirations are a key determinant of children's academic achievement, yet students may also be influenced by the aspirations of their classmates' parents. Using the China Education Panel Survey (CEPS) data and exploiting random classroom assignments, I assess the impact of peer parental aspirations on test scores. High peer parental aspirations significantly enhance scores, particularly when combined with high parental aspirations indicating a social multiplier in educational investment. These effects exhibit substantial heterogeneity, with larger and more persistent impacts for boys, while girls' outcomes depend on peer-family aspiration alignment, and effects are stronger in mathematics than in language subjects.

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² Yuzhi Zhou is the corresponding author. Research School of Economics, Australian National University, Kingsley Pl, Canberra, 2600, ACT, Australia. During the preparation of this work, Yuzhi Zhou used ChatGPT and Deepseek in order to refine language and assist editing. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

1 Introduction

Students' academic performance is significantly influenced by social interactions within classrooms. A rich body of literature has established that parental inputs, peer environments and peer family input independently affect student outcomes (i.e. Agostinelli et al., 2026; Chung, 2020; Chen et al., 2023). Among these extensive research has documented the impact of parental aspirations on individual student performance.³ However, little is known about how peer parental aspirations, for example, university ambitions that classmates' parents hold for their children, might create spillover effects that shape academic achievement across classrooms. Are these aspirational norms helpful for students? If so, will all students benefit?

This paper investigates whether and how peer parental aspirations shape students' academic performance, drawing on data from the China Education Panel Survey (CEPS). Although linear-in-means (LIM) models are popular in this area, recent research suggests that they may not capture the full complexity of peer interactions (i.e. Boucher et al., 2024). I contribute to this literature by showing that peer parental aspirations are a critical social input whose effect is not uniform but generating heterogeneous returns that can reproduce educational inequality across generations. Exploring the random assignment of students to classrooms, I identify the causal impact of being surrounded by classmates whose parents have high educational aspirations. This design, leveraged by prior studies (i.e. Chung and Zou, 2023), ensures robust identification of peer influences on academic outcomes.

The main results demonstrate that peer parental aspirations exert a significant positive influence on student achievement, with particularly strong effects when students' parents also hold high aspirations. This suggests that a high peer parental aspiration environment complements rather than substitutes family environment.

³ The term "Aspirations" rather than "Expectations" is used because aspirations more accurately capture parents' normative goals and long-term ambitions for their children, especially in early adolescence when educational paths remain fluid. While expectations reflect constrained predictions based on current performance or resources, aspirations signal what parents want their children to achieve and are thus more responsive to social norms and peer influence. In the context of CEPS, the phrasing of the survey question aligns with this interpretation, and the use of aspiration allows for a clearer examination of how high peer parental aspirations generate motivational spillovers and shape educational behavior within classrooms.

One potential explanation is that academically supportive peer environments may not fully offset disadvantages rooted in family background. Students from low-aspiration families may face increased academic competition in a high peer parental aspiration environment, potentially exacerbating achievement gaps.

In addition, the effects vary by subject and gender, with magnitudes comparable to baseline estimates. Notably, mathematics exhibits both the strongest complementarity and the most pronounced penalties from aspiration mismatches. While language subjects show more muted effects, particularly Chinese, which displays the weakest impact. Regarding gender differences, girls appear to be more sensitive to aspiration misalignment between family and peer contexts. While boys exhibit large and significant gains, independent of family aspiration.

To show robustness, the supplementary analysis restricts the sample to students surveyed immediately after classroom enrollment, minimizing the risk of reverse causality from peer exposure. Further checks accounting for peer parental education, excluding schools with reported parental influence in placement and outlier classrooms with extreme sizes also produce consistent results, reinforcing the credibility of the main findings.

This study identifies multiple channels. First, students directly internalize the educational norms from high-aspiration peer environments, raising their own academic ambitions. Second, these aspirations operate through peer channels by fostering more positive classroom climates and stronger academic peer norms. Third, parents in high-aspiration peer groups demonstrate increased academic supervision and engagement, particularly for students from less-aspirational family backgrounds. Fourth, while peer parental aspirations improve parental perceptions of teacher quality, they show limited impact on teacher-student interactions.

These findings demonstrate how educational outcomes are shaped not just by individual family inputs but through complex social ecosystems where peer and parental aspirations interact. The results reveal a social multiplier effect. High aspiration peer environments amplify the benefits for already-advantaged students while doing little to compensate for disadvantaged students. This has

critical implications for educational equity. Policies may need to simultaneously address peer composition and family support systems to tackle inequality.

2 Literature Review

It is well established that parental input significantly shapes children's outcomes (e.g., Becker and Tomes, 1986; Boonk et al., 2018; Cui et al., 2019; Dizon-Ross, 2019; Boucher et al., 2023). Among the different forms of parental involvement such as family support or communication with teachers, parental aspirations stand out as one of the strongest predictors of student success (i.e. Boonk et al., 2018). In India, higher parental aspirations have been shown to significantly increase the likelihood of private-school enrollment (i.e. Georgiadis et al., 2022), while Koshy et al. (2019) find a similar pattern in Australia, where parental aspirations drive higher education participation. Likewise, in China, Lai et al. (2022) confirm that parental educational aspirations positively influence children's attainment of higher education. Overall, high parental aspirations generally lead to better academic performance, though the degree of influence varies across racial and ethnic groups (i.e. Yamamoto and Holloway, 2010).

Additionally, children's perceptions of their parents' aspirations can significantly impact their academic performance (i.e. Scho'rner and Bittmann, 2023). Moreover, students' beliefs about their own ability to complete higher education are heavily influenced by both peers and parents (i.e. Kunz and Staub, 2020). Furthermore, parental beliefs about their children's abilities influence how much they invest in their education Guo and Qu (2022), which likely reinforces the effect of their expectation.

In parallel, existing literature robustly documents significant peer effects in education, (i.e. Calv'o-Armengol et al., 2009; Feld and Zo'litz, 2017; Xu et al., 2022; Feng et al., 2022; Chen et al., 2024). Exposure to high-achieving peers (i.e. Zhao et al., 2017), persistent peers (i.e. Zou, 2024), and peers of different gender compositions (i.e. Gong et al., 2019), have been shown to significantly influence academic performance. Carrell and Hoekstra (2010) establish that exposure to peers from violent families significantly decreases classmates' reading and math

test scores. Regarding heterogeneous populations, Hu (2018) identifies negative effects from internal migrant peers on local Chinese students' academic performance, potentially through learning environment deterioration or teacher adjustments. Conversely, Pan and Luo (2023) find no significant influence of special needs peers.

The influence of peers with different parental aspiration remains understudied.

To address this, recent studies have begun to explore how the characteristics of peer parents affect student outcomes directly and indirectly. For example, Chen et al. (2023) show that when parents encounter socially influential peer parents, they are prompted to become more engaged in their children's education, resulting in improved academic outcomes. Chung and Zou (2023) further highlight the spillover effects of peer mothers' education, demonstrating how these educational backgrounds positively influence student performance by shaping parenting practices. Additionally, Chung (2020) shows that peer parental education has gender-specific spillover effects on university attendance, where role models play a key part in motivating students. These studies indicate that peer parental influence establishes benchmarks within social networks, raising the overall academic performance by influencing a wide range of families across different socio-economic and cultural backgrounds.⁴

Despite these findings, little is known about the specific impact of peer parental aspirations on student performance and how this influence interacts with a student's own parental aspirations. This study fills that gap by analyzing the interconnected effects of peer and own parental aspirations on educational outcomes. The findings offer a more nuanced understanding of the educational environment, highlighting that students are embedded in networks where parents, peers, and teachers simultaneously shape academic performance. Importantly, this research raises awareness of the potential for reinforcing performance gaps when

⁴ Neighborhood effects further underscore the importance of social context, as advantaged environments foster higher aspirations and improved long-term outcomes. Chetty et al. (2016) find that early relocation to lower-poverty areas improves long-term outcomes, later Chetty and Hendren (2018) show that extended exposure to high-opportunity neighborhoods enhances earnings.

parents, particularly those in less advantaged settings, should update their aspirations in response to positive peer benchmarks.

3 Background

In China, compulsory education lasts for nine years, encompassing six years of primary education followed by three years of secondary schooling. Secondary education, which includes grades 7 to 9, is the final stage of mandatory schooling, after which students compete for placement in high schools through a selective entrance exam process. In 2022, approximately 58% of secondary school graduates were enrolled in traditional high schools, while the remaining students either opted for vocational training or discontinued formal education altogether.⁵

Generally, admission to secondary schools is determined by residential zoning regulations. Most students are assigned to schools based on their Hukou, a household registration system that ties individuals to a specific locality.⁶ Under this policy, students typically attend schools in their registered district and are not permitted to enroll in schools outside their designated area.⁷ On average, a secondary school accommodates students from multiple primary schools, which means that many students will meet new classmates when they enter seventh grade. According to government regulations, Chinese public secondary schools typically assign students to classrooms through random selection. This practice is intended to ensure fairness and prohibits schools from grouping students according to ability, background, or other criteria, and students generally remain in the same class until they complete secondary education.

⁵ According to the latest official statistics, the proportion of secondary school graduates enrolling into high school was about 58% in 2022, compared to 53% in 2013, when the initial wave of the China Education Panel Survey (CEPS) was conducted (see http://www.moe.gov.cn/jyb_sjzl/sjzl_fztjgb/202307/t20230705_1067278.html, in Chinese)

⁶ Hukou represents China's household registration system that designates individual residency status within specific administrative areas while simultaneously encoding extensive information about local community ties and geographic affiliations. It is a dual-classification framework that assigns residency status by both geography (specific cities/counties) and type (rural vs. urban).

⁷ While hukou-based school assignment could raise concerns about residential sorting, I use school-grade fixed effect to fully absorb time-invariant district-level selection.

The secondary school curriculum, standardized by the Ministry of Education, places strong emphasis on three subjects: Chinese, Mathematics, and English. Despite some reforms aimed at promoting more interactive learning and fostering critical thinking, instruction remains focused on academic performance and preparing students for exams. Assessment in secondary schools is highly test oriented, with regular quizzes, midterms, and finals serving as key indicators of student performance. These assessments are particularly important because they are strong predictors of the results of the High School Entrance Examination, which determines a student's academic pathway.

4 Data and Variables

The China Education Panel Survey (CEPS) is a nationally representative longitudinal survey which covers 112 secondary schools and 438 classrooms in 28 counties or districts. In each school, 2 classrooms were randomly selected for each grade. With detailed surveys on students, parents, teachers, and principals, it captures a wide array of student characteristics, parental backgrounds, and teacher behaviors, which are uniquely linked through individual identifiers. The dataset is structured to allow for detailed analysis of factors such as classroom dynamics, student performance, and family background, providing an ideal setting to study peer effects and parental spillover mechanisms.⁸

The Chinese secondary education system operates on a two-semester academic calendar spanning approximately 40 weeks annually. The Fall semester

⁸ CEPS is the first and largest nationally representative survey of junior secondary students in China, employing a stratified, multi-stage probability sampling design to ensure broad coverage across regions, school types, and urban-rural areas. Detailed sampling design of CEPS is provided in the CEPS website, see http://ceps.ruc.edu.cn/English/Documentation/Sampling_Design.html. CEPS provides unique identifiers for each student, classroom, and school. The CEPS dataset encompasses five different questionnaires, which were administered to students, parents, homeroom teachers, major subject teachers, and school principals across both waves of the survey.

commences September 1st and concludes in late January or early February, contingent upon Chinese New Year timing, while the Spring semester runs from February through June.⁹ As for the specific timeline of CEPS, students in the baseline sample were officially enrolled on 1st September 2013. The CEPS survey was implemented in two distinct phases for both Wave 1 and 2: the first batches were conducted during the fall semesters, and the second batch occurred in the spring semesters. The midterm exams in the first semesters, a key academic milestone often used for short-term outcome measurement, were recorded by the school instead of by self-report. The exams typically occur during weeks 8-10 of each semester. For the Fall semester, they are usually conducted in November, shortly after the beginning of the school year.

In 2013, 19,487 students in grades 7 and 9 were surveyed.¹⁰ I restrict the sample to grade 7 students in schools with random assignment of students and teachers to classrooms, allowing causal identification of peer effects through panel methods.¹¹ The final dataset consists of 3,003 students across 94 classrooms in 47 different schools. Moreover, a subsample is constructed with students in grade 7 who were surveyed right after the enrollment in 2013 fall. Since students and parents would have had minimal exposure to their new peer group, thereby limiting the possibility of reverse causality or peer-induced bias in reported aspirations or attitudes.

⁹ See, for example, https://szeb.sz.gov.cn/home/jyfw/fwxsjz/gzjy/zktzgg/content/post_11363007.html and <https://www.chinaeducationaltours.com/guide/article-chinese-school-calendar.html>

¹⁰ There were 10,279 in grade 7 and 9,208 in grade 9. The typical age in grade 7 is 13.

¹¹ Initially, I identify schools that implement random student assignments, as reported by school principals, ensuring that Grade 7 and Grade 9 students are not regrouped based on academic performance. Subsequently, I utilize homeroom teachers' responses to exclude any school-grade cohorts where students are assigned to classes based on ability. Observations with missing responses to key questions are removed to maintain data integrity. Classes with fewer than two mother respondents or two father respondents were excluded as they lack the gender diversity necessary to form a representative measure. This threshold ensures that the peer parental aspiration variable is free from potentially skewed perspectives from a single gender.

4.1 Timeline

Figure 1 visualizes this timeline, highlighting enrollment, timing of Wave 1 (Baseline) batches, midterm exams.¹² Wave 1 data collection began shortly after student enrollment, ensuring that baseline academic conditions reflect the initial post enrollment environment. Wave 2 followed up with the same students during the 2014–2015 academic year, approximately one year after the baseline survey. This wave was designed to measure changes in outcomes and contexts, making it valuable for analyzing medium-term dynamics. Wave 1 test scores measure short-term peer and environmental influences (accumulated during the initial semester), while Wave 2 captures cumulative academic impacts following a full year of sustained peer interactions and social dynamics.

¹² The survey period spanned across 2 semesters for both wave 1 and wave 2 with dates approximated per CEPS documentation.

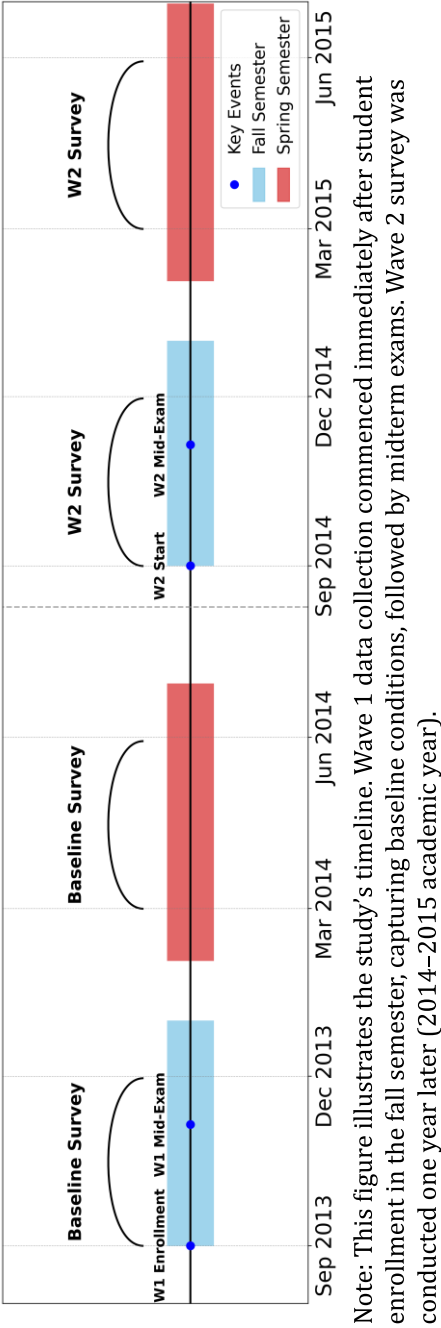


Figure 1: CEPS Timeline

Characteristic	Categories	2013 Fall (batch 1)		2014 Spring (batch 2)		Wave 1 All	
		Number	(%)	Number	(%)	Number	(%)
Student ID	Number of students surveyed	1,623	54.0	1,380	46.0	3,003	100.0
Class ID	Number of classes represented	52	55.3	42	44.7	94	100.0
School ID	Number of schools represented	26	55.3	21	44.7	47	100.0
County ID	Number of counties represented	14	58.3	10	41.7	24	100.0
School Location	Central area of city or county	774	47.7	676	49.0	1,450	48.3
	Urban fringe and rural-urban fringe	620	38.2	456	33.0	1,076	35.8
	Towns and rural area	229	14.1	248	18.0	477	15.9
School Type	Public school	1,594	98.2	1,205	87.3	2,799	93.2
	Private school	29	1.8	175	12.7	204	6.8
Family Wealth	Difficult	200	12.3	157	11.4	357	11.9
	Moderate	1,317	81.2	1,088	78.8	2,405	80.1
	Wealthy	105	6.5	134	9.7	239	8.0

Note: This table reports the student and school composition surveyed in two batches during Wave 1: Fall 2013 and Spring 2014. The Fall 2013 group (N = 1,623) was surveyed shortly after classroom enrollment, while the Spring 2014 group (N = 1,380) was surveyed later in the academic year. Distributions across school location, school type, and family wealth indicate strong comparability between the two batches. This comparability, combined with random classroom assignment, supports the validity of using the Fall 2013 subsample for robustness checks targeting peer exposure at the point of entry.

Table 1: W1 Student and School composition by Survey Period

Table 1 presents the demographic composition and sampling structure of the Wave 1 survey data, collected across two time periods: Fall 2013 and Spring 2014.¹³ The distribution shows strong comparability between the two batches across key characteristics including school location, school type, and family wealth. The table demonstrates that the sample maintains representative proportions across geographical, institutional, and socioeconomic dimensions throughout the survey period. Furthermore, the study's longitudinal design, tracking students across two time points, allows me to assess both immediate and longer-term peer

¹³ This consistency supports the methodological validity of using the Fall 2013 subsample for robustness checks examining peer exposure effects at the point of classroom entry.

effects. The presence of 2 classrooms per grade provides the necessary variation in peer groups to disentangle peer influences from other confounding factors.

4.2 Random Classroom Assignment in CEPS

CEPS includes 57% of schools that employ random assignment of students to classrooms in Grade 7, as confirmed by principal reports. As outlined by the Ministry of Education in China, students are usually randomly assigned to classrooms through computerized or manual processes, with the principle of randomization to promote equity in educational opportunities. This randomization balances student characteristics and enables causal analysis of peer effects. Wave 1 Parental aspirations were measured immediately after enrollment, with 91.8% expecting high school completion and 73.4% anticipating university attainment. The analysis uses only schools with verified random assignment to estimate peer parental aspiration effects before sustained peer interactions occur.

Building on this randomization approach, Zou (2024) and Chung and Zou (2023) study the effects of persistent peers and peer parental education on student performance, while Dai and Zhou (2025) analyze how in-class rank influences shadow education behavior. By utilizing randomization, I can confidently attribute differences in academic performance, and behavior to peer parental influences, rather than to any pre-existing differences among students.¹⁴

Initially, I identify schools that implement random or balanced student assignments, as reported by school principals, ensuring that students are not regrouped based on academic performance. Subsequently, I utilize homeroom teachers' responses to exclude any school-grade cohorts where students are assigned to classes based on ability. Observations with missing responses are removed to maintain data integrity. I then generate school-grade-class identifiers and exclude school grade levels with only one class to preserve variation in peer

¹⁴ The validity of this design is supported by balance tests for both randomized and nonrandomized samples, reported in Table 3.

composition.¹⁵ Finally, observations with missing responses for either waves or negative scores are dropped to ensure data quality. In the following section, a balancing test is conducted to verify the validity and effectiveness of the classroom randomization.

4.3 Variables

4.3.1 Peer Parental Aspiration

To construct the key variable, peer parental aspiration (hereafter “PPA”), I employ the “leave-me-out” methodology.¹⁶ In this study, I define high parental educational aspiration as parents expecting their child to attain a university degree or higher. This is based on responses to the question: “What is the highest level of education you expect this child to receive?” The responses range across nine levels, from “Drop out now” to “Get a Doctor degree”. Similar question was used to construct Student Aspiration.

As shown in Figure 2, there is wide distribution of student test scores at every level of wave 1 parental aspiration. This pattern suggests that at a point when peer interactions had not yet formed, parental aspiration functions less as an expectation based on a rational assessment of the student’s ability and more as a long-term ambition variable. This provides compelling evidence that parental aspiration may not be not a proxy for private information about a child’s latent ability. The *PeerParAsp_{ics}* variable is constructed by calculating the leave-me-out average of this binary indicator within each class, as shown by the following formula:

$$PeerParAsp_{ics} = \frac{\sum_{j \in c} HighParAsp_j - HighParAsp_i}{NumParRes_c - 1}.$$

¹⁵ Missing data (1,322 of 4,325 observations) exhibits slightly lower peer parental aspirations, but the small magnitude (difference = 0.011) suggests minimal bias in the analysis. It suggests any bias in the estimated peer effect is likely smaller than the estimate. T-test is provided in the Appendix.

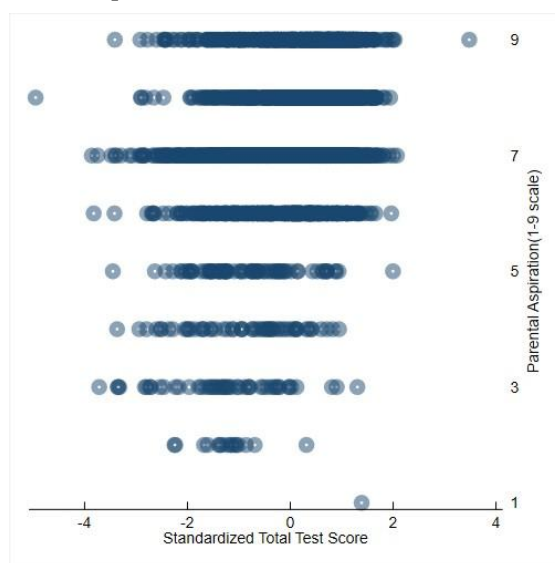
¹⁶ This approach is prevalent in educational studies utilizing CEPS dataset. For instance, Chung (2020) and Chung and Zou (2023) used it to construct a peer parental education variable. Similarly, Chen et al. (2023) applied the method to estimate the proportion of peers with cadre parents, and Huang and Zhu (2020) estimated the share of repeaters among peers.

$HighParAsp_i$ is a binary indicator equal to 1 if the parents of student i in classroom c of School-Grade group s hold high educational aspirations (expect children to attend university or above), and 0 otherwise.¹⁷ $\sum_{j \in c} HighParAsp_j$

is the total number of parents j with high educational aspirations in classroom c .

$NumParRes_c$ is the total number of parent respondents in classroom c .

The calculation excludes the focal student's own parental aspiration indicator from the classroom sum and is divided by the number of remaining parents in the classroom. Hence, $PeerParAsp_{ics}$ represents the average level of parental aspirations among all other parents in student i 's classroom.



Note: The scatter plot shows little correlation between Parental Aspiration and Student Scores. There is wide distribution of student test scores at every level of parental aspiration, suggesting that parental aspiration functions less as a rational assessment of the student's measured ability and more as an ambition variable.

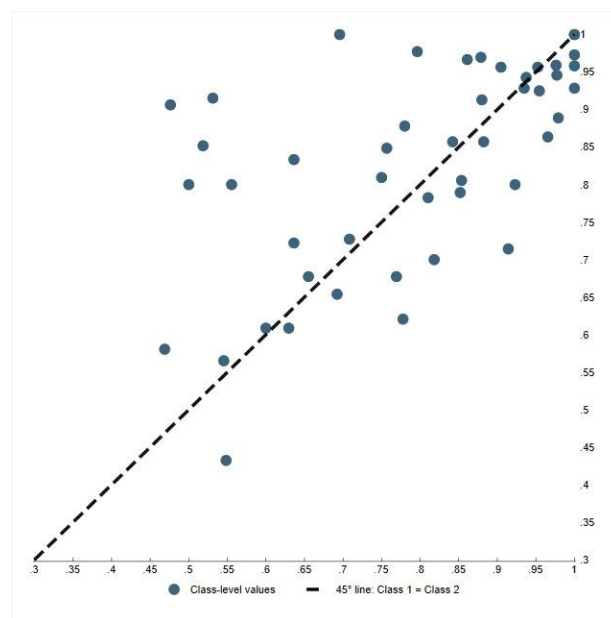
Figure 2: Parental Aspiration and Student Ability

Figure 3 presents a comparison of the average parental aspirations between Class 1 and Class 2 within same grade across schools in Wave 1.¹⁸ Each point

¹⁷ The distribution of classroom-level parental aspiration proportions is presented in Appendix Figure A1.

¹⁸ In each school, two classrooms were randomly selected per grade.

represents a school-grade, plotting the class-level averages for the two randomly selected Grade 7 classrooms. The 45-degree dashed line serves as a reference point where Class 1 and Class 2 would have identical average parental aspirations. The dispersion of points around this line clearly indicates substantial cross-classroom variation within schools. Importantly, while many points lie near the 45-degree line, suggesting some degree of similarity, there are numerous instances of meaningful deviation, especially at both ends of the aspiration distribution. This variation is not only statistically notable but also crucial for identification in empirical work. Because students were randomly assigned to classrooms, the divergence in class-level parental aspiration reflects quasi-experimental variation.



Note: This figure compares average parental aspirations between Class 1 and Class 2 within each grade in Wave 1. Each point represents a grade, with the axes denoting class-level averages. Deviations from the 45-degree line reflect variation in classroom-level parental aspirations, which provides quasi-experimental leverage for identifying peer effects.

Figure 3: Comparison of Within Grade Classroom Level Parental Aspiration

4.3.2 Student's Test Score

The CEPS collected midterm scores in Chinese, Mathematics, and English from school archives.¹⁹ To ensure comparability, scores were standardized at the school grade level. While midterm exam scores are directly comparable within the same grade and school, the standardized curriculum set by the Education Ministry provides some consistency across schools, allowing for partial comparability within the same grade. This study uses the midterm scores to measure student achievement, as high school admission is based on a composite test score across subjects.

4.3.3 Controls

A comprehensive set of student, peer, and teacher covariates are incorporated to address heterogeneity across individuals and classrooms to improve estimation precision and strengthen causal identification of peer parental aspiration effects.

Student controls: At the student level, the analysis incorporates a wide range of predetermined characteristics and early educational experiences. Demographic controls include age, gender, and ethnicity, along with residential (rural or urban) and migration indicators. Family background controls capture the number of siblings and any early-childhood participation in kindergarten. Additional school history variables include the student's age at the time of primary school enrollment and whether the student skipped or repeated a primary school grade. To further account for differences in behavior and attitude, the model controls for key non-cognitive traits, such as perseverance and other non-cognitive abilities.²⁰

¹⁹ In China, an academic year has two semesters of approximately 20 weeks each, with midterms typically held between weeks 8 and 10.

²⁰ In the CEPS wave 1, students were asked about seven questions related to their retrospective personality traits in grade 6. These variables are constructed on students' responses to seven retrospective questions about personality traits in Grade 6.

Following Gong et al. (2018), Chung (2020), Chung and Zou (2023), and Zou (2024), I construct a *Perseverance Index* using the related questions and create an *Other Non-Cognitive Traits Index* using related questions.²¹

Peer controls: To account for classroom-peer effects, I employ leave-me-out averages of classmates' characteristics, to prevent mechanical correlations between individual attributes and peer measures in predicting student outcomes.²² Following previous literature Gong et al. (2019), Hu (2018), Xu et al. (2022) and Zou (2024), I control classmates' gender, demographic profiles, migration status, academic history, persistence and other non-cognitive ability. This classroom-level measurement approach, feasible due to random classroom assignment, enables more precise isolation of peer environmental effects in the empirical framework.

Teacher controls: The final set of controls addresses variation in teaching quality and style with homeroom teacher characteristics, such as demographic attributes (age, gender, marital status), professional qualifications (educational attainment, professional certification, pedagogical degree graduation status), and years of teaching experience. This helps isolate the effects by accounting for systematic differences in teaching approaches and effectiveness across classrooms. Additionally, controls for subject teachers in Math, English, and Chinese were used in the robustness checks.

²¹ The *Perseverance Index* is constructed from student responses to three items regarding their willingness to attend school despite discomfort, complete disliked homework, and persist with lengthy assignments (4 scale: Strongly Disagree to Strongly Agree). The *Other Non-Cognitive Traits Index* aggregates responses to four questions on self-expression, response quickness, learning speed, and curiosity. Both indices are created using a two-step standardization procedure: first, individual item responses are standardized and averaged within each index; second, this average is standardized across the estimation sample.

²² In line with the methodology proposed by Altonji and Mansfield (2018), this method is a common strategy to disentangle true peer effects from mechanical correlations and is adopted by many related studies (i.e. Booij et al., 2017; Chung and Zou, 2023; Z'arate, 2023).

4.4 Summary Statistics

This study leverages Wave 1 survey which is conducted immediately after secondary school enrollment, so that pre-treatment measures are plausibly exogenous to peer influence, reducing concerns about reflection bias and endogeneity. This timing is crucial for identification: by capturing aspirations before peer affects each other, I treat them as predetermined rather than outcomes of social processes. For academic outcomes, Wave 1 test scores establish baseline achievement while Wave 2 scores measure the sustained impact.

Table 2 presents detailed summary statistics across four analytical dimensions. Panel A reports the key outcome and treatment variables: both Wave 1 and Wave 2 test scores are standardized to facilitate interpretation, and 83% of students have parents who aspire for their university attendance. Panel B characterizes the student sample, with mean age of 13.42 years, 50% female, 8% ethnic minority representation, and 38% holding agricultural hukou status. Panel C displays peer characteristics computed as within-classroom leave-me-out means. Panel D documents homeroom teacher characteristics: teachers average 36.77 years of age, are predominantly female (75%), hold pedagogical degrees (94%), and possess universal teaching certification, indicating high professional qualification levels across the sample schools.

	Mean	SD	Min	Max
A. Key variables				
W1 Total test score (Standardized)	0.02	0.98	-4.95	3.48
W2 Total test score (Standardized)	0.00	0.99	-5.91	2.98
Parental High Aspiration	0.83	0.37	0.00	1.00
Peer parental aspiration	0.83	0.15	0.41	1.00
B. Individual characteristics				
Age	13.42	0.61	12.00	16.00
Gender (Female = 1)	0.50	0.50	0.00	1.00
Ethnicity (Minority = 1)	0.08	0.27	0.00	1.00
Hukou status (Agricultural = 1)	0.38	0.49	0.00	1.00
Migrant status (Migrant =1)	0.21	0.41	0.00	1.00
Number of siblings	0.51	0.71	0.00	5.00
Attend kindergarten	0.86	0.35	0.00	1.00
Age attending primary school	6.67	0.95	0.00	9.00

Repeat grade in primary school	0.08	0.27	0.00	1.00
Skip grade in primary school	0.01	0.11	0.00	1.00
Persistence in Grade 6 (Standardized)	0.18	0.92	-3.41	0.95
Non-cognitive measure in Grade 6 (Standardized)	0.11	0.95	-3.64	1.31
C. Peer characteristics				
Peer: Gender (Female = 1)	0.50	0.09	0.00	0.83
Peer: Migrant status (Migrant =1)	0.23	0.21	0.00	1.00
Peer: Repeat grade in primary school	0.08	0.12	0.00	0.70
Peer: Persistence in Grade 6	0.15	0.22	-0.71	0.77
Peer: Non-cognitive measure in Grade 6	0.08	0.21	-0.46	0.53
D. Homeroom teacher characteristics				
Age	36.77	6.86	23.00	60.00
Gender (Female = 1)	0.75	0.43	0.00	1.00
Marriage status	0.10	0.31	0.00	1.00
Has a university degree	0.55	0.50	0.00	1.00
Teaching experience in years	14.79	7.62	1.00	38.00
Graduated from a pedagogical degree	0.94	0.24	0.00	1.00
Teacher certificate	1.00	0.00	1.00	1.00
Number of observations	3003			

Note: The sample consists of students with random classroom and teacher assignment. Panel A presents the primary outcomes: Wave 1 and 2 total scores (sum of Chinese, Math, and English), a binary indicator for high parental aspirations, and the leave-me-out average of peer high parental aspirations. Panels B–D summarize student, peer, and teacher controls.

Table 2: Summary Statistics of Working Sample

5 Estimation Strategy

This study estimate classroom peer effects while controlling multiple sources of unobserved heterogeneity that may confound the relationship between peer parental aspirations and student outcomes. Following Burke and Sass (2013), I implement a fixed-effect model to account for time-invariant school and instructor characteristics that influence academic performance. This approach is consistent with evidence from Chetty et al. (2011), who demonstrate that teacher quality effects are both substantial and persistent across student cohorts.

Selection Biases: China’s education system is known for the fierce competition. Parents often strive to enroll their children in top secondary schools, while prestigious institutions seeking to sustain their reputations, admit only the strongest applicants. Such mutual incentives can give rise to self-selection bias,

whereby students are grouped with peers of similar academic ability and potentially other unobserved characteristics. Moreover, when parents select specific schools or classes based on their aspirations, the resulting non-random sorting can bias empirical estimates by capturing parental selection rather than true effects.

For sorting in school selection, most students attend schools assigned according to their residential zones in China. With school-grade fixed effects, school selection could be addressed.²³ Here, I also rely on the random assignment, which ensures that classroom composition is unrelated to parental preferences.

The Reflection Problem: To address the potential reflection problem, I have adopted two primary strategies. First, I construct the peer parental aspiration variable using a leave-me-out approach. As mentioned above and shown in Figure 2, parental aspirations capture long-term ambitions, which are unlikely to be influenced by short-term academic fluctuations. Second, I use Wave 1 peer parental aspirations as the key independent variable in both waves' analyses. In the case of Wave 2, this allows for a lagged model specification, in which outcomes are regressed on peer characteristics measured prior to them. For Wave 1, aspirations were measured before meaningful peer interactions had time to develop.²⁴

Endogeneity of Parental Aspirations: Endogeneity is often a central concern when estimating the effect of parental aspirations on student outcomes, as noted by Georgiadis et al. (2022). If aspirations are influenced by a child's intrinsic ability, unobserved family resources, or prior performance, a simple regression of student achievement on parental aspirations can lead to biased estimates. To address the endogeneity of parental aspirations, I leverage random classroom assignments for exogenous peer group formation and exploit the timing of the survey, fielded immediately after enrollment to exclude peer-influenced beliefs.²⁵ This strategy provides a valid causal estimate of peer parental aspiration effects.

²³ Angrist et al. (2013) also strengthen the importance of accounting for school-level differences.

²⁴ As an additional robustness check, I analyze a subsample of students surveyed immediately after school entry, further supporting the exogeneity of Wave 1 aspirations.

²⁵ To further isolate the effect of aspirations from bundled parental education, robustness check controlling for parental education was performed.

Robust causal inference regarding the effect of peer parental aspirations must overcome three principal empirical challenges: selection biases (i.e. Angrist, 2014), the reflection problem (i.e. Manski, 1993), and the endogeneity of parental aspirations. While random assignments eliminate self-selection and the reflection problem, it does not isolate parental aspirations from other correlated traits. The following section presents balance tests verifying that randomization successfully equalized observable peer characteristics across classrooms. Table A3 further tests whether the estimated peer parental aspiration effect operates independently of correlated socioeconomic attributes by augmenting the baseline specification with controls for peer parents' educational attainment and family wealth.

5.1 Balancing Test

The randomization approach was confirmed in prior studies (i.e. Zou, 2024; Chung and Zou, 2023). Here, I compare balancing tests with two samples: one incorporates randomization and another that excludes it. Table 3 presents results from balancing tests to examine whether predetermined student characteristics predict the leave-me-out Wave 1 Peer Parental Aspiration variable, providing evidence of the validity of the randomization. The balancing test includes regressions of student characteristics on PPA, comparing results across specifications.

The non-randomized sample reveals systematic sorting, with OLS results (Cols 1-2) showing significant correlations between student characteristics and peer parental aspirations. While school-grade fixed effects (Col 3) attenuate most associations, residual links with ethnicity, hukou status, and non-cognitive skills persist, highlighting the limitations of fixed effects alone in addressing non-random assignment.

The randomized sample (Cols 4-6) demonstrates successful experimental design: initial correlations between student traits and peer aspirations (Cols 4-5) are fully absorbed by school-grade fixed effects (Col 6), confirming within-school randomization validity within school.

Peer Parental Aspiration	Without Randomization			With Randomization		
	OLS (1)	OLS (Clustered SE) (2)	Fixed Effects (3)	OLS (4)	OLS (Clustered SE) (5)	Fixed Effects (6)
Age	-0.014*** (0.004)	-0.014* (0.007)	-0.003 (0.002)	-0.005 (0.005)	-0.005 (0.006)	0.000 (0.003)
Gender (Female)	0.003 (0.005)	0.003 (0.004)	0.002 (0.002)	0.012** (0.005)	0.012** (0.005)	0.005 (0.003)
Minority	-0.010 (0.009)	-0.010 (0.008)	-0.023** (0.011)	-0.019* (0.009)	-0.019 (0.025)	0.009 (0.007)
Hukou (Agricultural)	-0.054*** (0.005)	-0.054*** (0.014)	-0.008** (0.003)	-0.098*** (0.006)	-0.098*** (0.021)	-0.010 (0.008)
Migrant	-0.006 (0.006)	-0.006 (0.025)	-0.012** (0.005)	-0.002 (0.006)	-0.002 (0.022)	0.004 (0.006)
Number of Siblings	-0.047*** (0.003)	-0.047*** (0.008)	0.001 (0.002)	-0.024*** (0.004)	-0.024*** (0.008)	-0.003 (0.005)
Kindergarten Attendance	0.013** (0.006)	0.013 (0.012)	0.002 (0.004)	0.042*** (0.007)	0.042*** (0.011)	0.012 (0.008)
Age attending Primary school	0.006*** (0.002)	0.006* (0.005)	0.001 (0.001)	-0.004 (0.003)	-0.004 (0.010)	0.002 (0.002)
Skip grade in Primary school	-0.030 (0.021)	-0.030 (0.022)	-0.014 (0.013)	-0.010 (0.023)	-0.010 (0.028)	0.003 (0.009)
Repeat grade in Primary school	-0.046*** (0.007)	-0.046*** (0.016)	0.005 (0.005)	-0.042*** (0.010)	-0.042** (0.019)	-0.012 (0.011)
Perseverance (SD)	-0.001 (0.003)	-0.001 (0.004)	-0.002 (0.001)	-0.007** (0.003)	-0.007 (0.005)	0.003 (0.002)
Non-Cognitive Skills (SD)	0.009*** (0.003)	0.009** (0.004)	0.003* (0.001)	0.008** (0.003)	0.008 (0.005)	0.000 (0.002)
Constant	0.985*** (0.048)	0.985*** (0.103)	0.807*** (0.031)	0.932*** (0.061)	0.932*** (0.096)	0.807*** (0.042)
Observations	3,313	3,313	3,313	3,003	3,003	3,003
R-squared	0.2295	0.2295	0.8075	0.1908	0.1908	0.7908
F-stat	81.89***	11.12***	1.21	58.75***	6.58***	1.07
SchGradeFE	N	N	Y	N	N	Y

Note: This table presents balancing tests examining whether predetermined student characteristics predict the leave-one-out Wave 1 peer parental aspiration measure. Cols (1)-(3) report results for the sample without random classroom assignment, while cols (4)-(6) show results for the sample with randomization. Specifications include ordinary least squares (OLS) regressions with and without clustered standard errors, as well as models with school grade fixed effects (SchGradeFE). The tests assess the validity of the randomization by evaluating the association between predetermined student characteristics and peer parental aspirations. Robust standard errors are reported in parentheses; *: $p < 0.1$, **: $p < 0.05$, ***: $p < 0.01$.

Table 3: Balancing Test Results: Comparison by Randomization Status

The contrast in F-statistics and coefficient patterns across samples reinforces both the quality of the CEPS randomization and the necessity of fixed effects for addressing even minor baseline imbalances.²⁶

²⁶ As emphasized by Athey and Imbens (2017), randomization ensures identification but does not preclude finite-sample imbalances, making regression adjustments such as fixed effects both appropriate and efficiency-enhancing.

5.2 Econometric Specification

$$Total\ score_{itcs} = \alpha + \beta_1 PeerParAsp_{ics} + \beta_2 HighParAsp_i + X\gamma + \phi_{SchGradeFE} + \epsilon_{itcs}$$

Where:

Total score_{itcs}: Denotes the standardized test score for student *i* in SchoolGrade group *s* classroom *c* in Wave *t*.

PeerParAsp_{ics}: Leave-me-out average of Parental Aspirations (binary) for students in the same classroom *c* in School-Grade group *s*, excluding student *i*'s own parent in Wave 1.

HighParAsp_i: Binary indicator for whether student *i*'s parents have high educational aspirations for their child in Wave 1.

X: Vector of control variables, including students' own aspiration, student demographics, peer characteristics, and teacher attributes.

$\phi_{SchGradeFE}$: School-Grade fixed effects to account for contextual factors and sorting of students across schools.

ϵ_{itcs} : The idiosyncratic error term.

Including students' own aspirations in *X* is important when the outcome is test performance, as it helps separate the channel and ensures that β_1 reflects the influence of the peer parental environment net of individual ambition. A potential concern is that including it could introduce endogeneity if these were shaped by the peer environment. In this setting, however, aspirations are predetermined: they capture long-term goals (e.g., attending university) and were measured immediately after classroom assignment.

The inclusion of School-Grade fixed effects controls for all time-invariant characteristics at the School-Grade level, including institutional factors such as school quality, administrative policies, staff practices and local socioeconomic conditions. Additionally, this specification accounts for systematic sorting of students across schools and grades, thereby mitigating potential bias from omitted variables correlated with both peer composition and student outcomes. Including $\phi_{SchGradeFE}$ ensures that identification of peer effects comes from within School-Grade variation in Peer Parental Aspirations.

6 Main Results

	W1TotalScores			
	(1)	(2)	(3)	(4)
Peer High Parental Aspiration	1.204** (0.534)	0.889* (0.489)	0.709* (0.400)	0.972** (0.463)
Parental High Aspiration	0.480*** (0.064)	0.441*** (0.063)	0.444*** (0.060)	0.452*** (0.060)
Student Aspiration	0.589*** (0.048)	0.476*** (0.044)	0.457*** (0.043)	0.461*** (0.042)
	W2 Total Scores			
	(1)	(2)	(3)	(4)
Peer High Parental Aspiration	1.757*** (0.508)	1.459*** (0.470)	1.037** (0.432)	1.145** (0.521)
Parental High Aspiration	0.484*** (0.061)	0.451*** (0.061)	0.447*** (0.061)	0.450*** (0.062)
Student Aspiration	0.612*** (0.045)	0.496*** (0.043)	0.478*** (0.043)	0.479*** (0.043)
School-grade fixed effect	Y	Y	Y	Y
Student controls	N	Y	Y	Y
Peer controls	N	N	Y	Y
Teacher controls	N	N	N	Y
Observations	3003	3003	3003	3003

Note: This table reports ordinary least squares (OLS) estimates of the effect of peer parental aspirations on students' standardized test scores in Wave 1 & 2. Cols 2 through 4 progressively incorporate controls for student demographics, peer composition, and teacher characteristics. Robust standard errors clustered at the school-grade level are shown in parentheses; *: $p < 0.1$, **: $p < 0.05$, ***: $p < 0.01$.

Table 4: Baseline Impact of Peer Parental Aspirations on Test Scores

Table 4 presents a sequential regression analysis revealing robust and persistent effects of peer parental aspirations on student achievement, while progressively

controlling for confounding factors.²⁷ The most striking finding is the substantial peer effect that remains significant even after comprehensive controls for student demographics, peer composition, and teacher characteristics. In Wave 1, peer parental aspirations show a significant positive effect 0.972** which is associated with a jump from being an average student to being in the top 17% in the grade. in the fully specified model, while Wave 2 shows stronger persistence 1.145** which means a jump from average to the top 13%, indicating that exposure to high-aspiration peer environments confers lasting academic benefits.

The attenuation pattern across specifications reveals important mediating mechanisms. Big reductions occur when student controls are added (26% decrease in W1, 17% in W2), revealing substantial compositional confounding. To address the concern that the PPA effect might simply reflect standard peer effects, I explicitly control for a comprehensive set of peer-level variables, including peer ability, repetition rates, and demographic composition. Notably, peer controls drive the largest single reduction in Wave 2 (29% decrease), indicating that peer group composition explains nearly one-third of the sustained effect. This pattern suggests the transmission mechanism operates primarily through peer characteristics and the persistence demonstrates that PPA captures a unique social mechanism beyond conventional peer ability effects. The modest rebound when adding teacher controls (10% in W2) implies teachers could have larger impacts in low parental aspiration classrooms where students benefit more from additional support while showing diminishing returns in high parental aspiration peer groups.

Parental aspirations maintain remarkably consistent effects across all specifications (0.452*** W1, 0.450*** W2), underscoring family inputs as primary achievement drivers. The stability of these coefficients despite progressive control

²⁷ The analysis follows rigorous causal inference standards. The observed attenuation of peer effects when adding pre-specified controls reflects proper adjustment for confounding pathways, as emphasized by Gelbach (2016). This pattern aligns with Rosenbaum (2002) principle that pretreatment covariates improve precision without introducing bias in randomized designs. With limited observation number, the estimate may still have concerns about sample size raised by Freedman (2008). The stable and significant peer effect across specifications demonstrates robustness to alternative explanations while providing a conservative lower bound for peer influence.

inclusion suggests direct parental influences operate largely independently of the measured factors. The analysis demonstrates that while both peer and parental aspirations matter, they represent distinct pathways: PPA effects operate through environmental exposure and social learning mechanisms, while parental effects reflect more direct effect potentially through parental input.

The substantially larger magnitude of PPA effects, more than double that of parental or student aspiration, potentially reflecting a social multiplier mechanism: class-wide norms generate amplification effects through collective standard-setting, thereby enhancing impact. This pattern aligns with social interaction theory where network effects and conformity pressures yield non-linear returns from group composition (i.e. Sacerdote, 2011; Carrell et al., 2013; Calvó-Armengol et al., 2009), demonstrating that peer environments function as enabling platforms. Such settings not only replicate familial educational investment but amplify its effects.

These results highlight PPA has remarkable persistence. Even after accounting for all controls, exposure to a high PPA environment predicts substantial long-term gains (1.145**). In the Appendix, Table A3 showed the estimated effect of peer parental aspiration remains statistically significant and economically meaningful after controlling for peer parents' educational attainment and family wealth, confirming the robustness of this finding. The persistence of PPA effects, with stronger coefficients in the later assessment, challenges conventional wisdom about the fading of environmental influences. Instead, it suggests that early exposure to high aspiration peer environments creates self-reinforcing benefits that compound over time, possibly through improved academic behaviors, heightened expectations, or access to better educational resources.²⁸ Together, these findings suggest that early academically supportive environments generate self-reinforcing benefits that persist well beyond the initial period of exposure.²⁹

²⁸ It coincides with the finding of Agostinelli et al. (2026) that a favorable environment positively affects students' development.

²⁹ Models without "Student aspiration" are provided in Appendix, and the results are robust.

7 Heterogeneous Effects

7.1 Family Aspiration

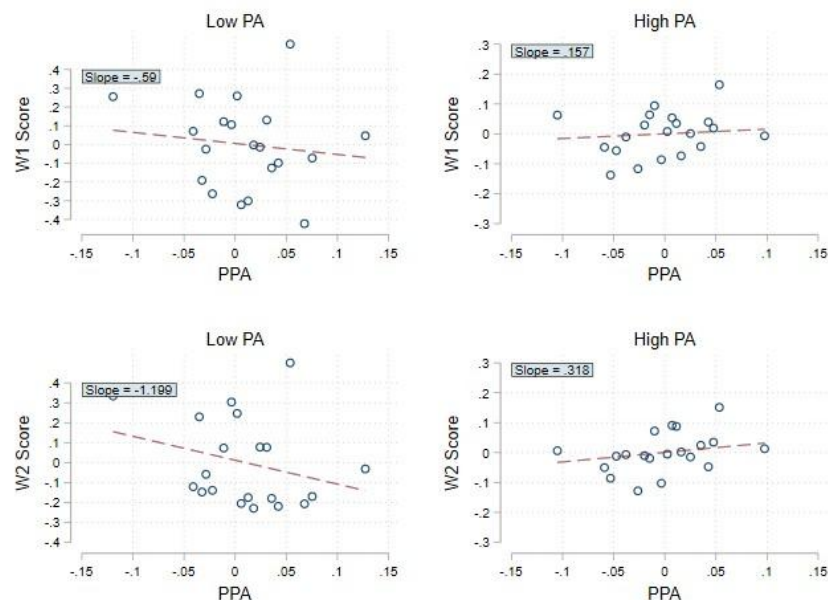
The findings in Table 4 establish the significant lagged influence of peer parental aspirations on student achievement, with effects persisting across increasingly demanding specifications. While these average effects are informative, they may mask important variations across different student populations, a common limitation in peer effects research. As shown by Boucher et al. (2023), children's educational outcomes are influenced by the traits of their families. Here, the economic magnitude of these effects warrants deeper investigation into potential heterogeneity, particularly given the theoretical ambiguity about whether peer parental aspirations might serve as substitutes or complements to students' own family inputs. CEPS's longitudinal design, leveraging random classroom assignments in the first place, provides clean identification to investigate this concern.

A fundamental question emerges: do all students benefit equally from exposure to peers with high-aspiration parents, or do these influences vary systematically with students' own parental aspirations? Standard regression models with additive specifications cannot fully address this question. Therefore, I employ bin scatter plots to visually examine the relationship between peer parental aspirations and academic achievement, separately by students' own parental aspiration status. This approach allows for the detection of interaction effects that might be obscured in aggregate analyses, providing visual evidence to guide more complex model specifications incorporating interaction terms.

There are theoretical reasons to expect heterogeneous effects. For example, a student from a high-aspiration family may especially benefit from classmates whose parents also have high aspirations, because the two environments reinforce each other's encouragement of effort and achievement (i.e. Zhao and Zhao, 2022). In other words, strong family expectations can motivate a student to capitalize on a motivating, goal-oriented peer group. By contrast, a student from a low-aspiration family may experience pressure when surrounded by families with

much higher goals. They could feel alienated or pressured by an environment that highlights the gap between their own circumstances and those of their peers.

Figure 4 reveals an intriguing pattern of heterogeneity in peer effects across students with different parental aspiration levels. In both waves, students whose parents have low university aspirations display a negative relationship between peer parental aspirations and academic performance, while those with high-aspiration parents show a slightly positive association. This visual evidence suggests that peer influences may operate differently depending on a student's family background, with potentially complementary effects for students already benefiting from high parental aspirations and substitution effects for those lacking such home advantage.



Note: This figure presents residualized binned scatter plots, conditional on all covariates and fixed effects. The results for Wave 1 confirm heterogeneous peer effects, with negative relationships for students of low-aspiration parents and a positive relationship for students of high-aspiration parents. As for Wave 2, there was persistent heterogeneous peer effects.

Figure 4: PPA Effects on Score by High VS Low Parental Aspiration

7.1.1 Interaction Model

To empirically capture the differential responses shown in Figure 4, it is essential to include an interaction term between peer and parental aspiration in the regression model as shown below.

$$\begin{aligned} Total\ score_{itcs} = & \alpha + \beta_1 PeerParAsp_{ics} + \beta_2 HighParAsp_i \\ & + \beta_3 (PeerParAsp_{ics} \times HighParAsp_i) + X'\gamma + \phi_{SchGradeFE} + \epsilon_{itcs} \end{aligned}$$

This allows the analysis to test whether the academic benefit of high peer parental aspirations is enhanced or diminished depending on the student's own parental aspiration level, thereby shedding light on the potentially complementary or contrasting roles of family and peer parental contexts in shaping student outcomes.

	W1TotalScores				
	(1)	(2)	(3)	(4)	(5)
Peer High Parental Aspiration	1.204** (0.534)	0.210 (0.691)	-0.185 (0.625)	-0.373 (0.507)	-0.095 (0.537)
Parental High Aspiration	0.480*** (0.064)	-0.454 (0.375)	-0.569 (0.343)	-0.544 (0.319)	-0.517 (0.314)
Student Aspiration	0.589*** (0.048)	0.592*** (0.047)	0.479*** (0.044)	0.461*** (0.043)	0.465*** (0.041)
Peer High Parental Aspiration × Parental High Aspiration	–	1.250** (0.484)	1.352*** (0.443)	1.321*** (0.419)	1.294*** (0.414)
	W2 Total Scores				
	(1)	(2)	(3)	(4)	(5)
Peer High Parental Aspiration	1.757*** (0.508)	0.995 (0.661)	0.619 (0.615)	0.177 (0.572)	0.278 (0.638)
Parental High Aspiration	0.484*** (0.061)	-0.232 (0.353)	-0.338 (0.326)	-0.338 (0.314)	-0.336 (0.308)
Student Aspiration	0.612*** (0.045)	0.614*** (0.045)	0.499*** (0.043)	0.481*** (0.043)	0.482*** (0.043)

Peer High Parental Aspiration × Parental High Aspiration	–	0.958** (0.476)	1.056** (0.445)	1.049** (0.433)	1.050** (0.425)
School-grade fixed effect	Y	Y	Y	Y	Y
Student controls	N	N	Y	Y	Y
Peer controls	N	N	N	Y	Y
Teacher controls	N	N	N	N	Y
Observations	3003	3003	3003	3003	3003

Note: This table explores whether the impact of peer parental aspirations on academic performance varies by students' own parental aspirations. The inclusion of an interaction term allows for the assessment of heterogeneous peer parental effects by parental aspiration. Robust standard errors clustered at the school-grade level are shown in parentheses; *: $p < 0.1$, **: $p < 0.05$, ***: $p < 0.01$.

Table 5: Interaction Models of Peer Parental and Parental Aspirations

Table 5 presents models that include the interaction between peer parental and own parental aspirations. In these models, the interaction term is not only positive and statistically significant but also demonstrates notable stability across specifications. This implies that the positive spillover of peer high parental aspirations is amplified when combined with high parental aspirations from the student's own family. Economic interpretation suggests important network externalities in human capital formation.

However, introducing the interaction term in Columns 2-5 causes the main peer effect to attenuate substantially and lose statistical significance, while the parental aspiration coefficient turns negative though remains largely insignificant. This pattern indicates that peer parental aspirations generate heterogeneous treatment effects based on family background: students with high-aspiration parents experience strongly positive combined effects (1.294*** in W1 and 1.050** in W2), while those with low-aspiration parents face negative net effects in W1 (-0.095) and negligible effect in W2 (0.278). This pattern suggests two potential mechanisms: (1) academically supportive peer environments may be insufficient to overcome disadvantages rooted in family background initially, or (2) these students may face heightened academic competition when first placed in high-aspiration peer groups, potentially exacerbating achievement gaps.

The robustness of interaction effects across both survey waves provides evidence of persistent complementarity in aspiration effects. Wave 2 estimates demonstrate that the long-term benefits of high-aspiration peer parental environments are concentrated among students who also come from high-aspiration families. This evidence is consistent with a complementarity mechanism in educational production, where peer and family inputs function as complements rather than substitutes in human capital formation. Policies that improve average peer quality without addressing heterogeneity in family background could potentially widen achievement gaps, as the benefits of enhanced peer environments accrue disproportionately to students already advantaged by supportive family contexts.

7.2 Subject

Prior research suggests that peer effects may vary across academic subjects, with Zimmerman (2003) finding stronger peer effects for verbal SAT performance compared to mathematics. Given this potential heterogeneity, I examine whether peer parental aspirations differentially influence performance across subjects. While the baseline analysis employs total test scores and controls for homeroom teacher characteristics, I re-estimate the model using subject-specific scores and incorporating subject-specific teacher controls to capture the distinct influence of specialized instructors on student outcomes.

Table 6 demonstrates heterogeneous effects between peer and parental aspirations across subject-specific assessments, with magnitudes comparable to baseline estimates, confirming these social spillovers changes subject-specifically. Striking disciplinary heterogeneity emerges: mathematics shows both the strongest complementarity and most severe penalties for aspiration mismatches, consistent with its role as a high-stakes screening subject where social and familial inputs compound, consistent with previous research results on peer effect shown by Carman and Zhang (2012). In contrast, language subjects exhibit attenuated effects (with Chinese showing the weakest effect), reflecting fundamental differences in skill acquisition. The weaker effects on language subjects may stem

from their more cumulative skill development processes and lower competitive pressures in standardized tracking systems. The significantly larger effect in quantitative curricula underscores that the returns to family-society aspiration alignment are not uniform. This points to a strategic policy priority: interventions in STEM fields, where early gaps compound, offer the greatest potential for impacting human capital accumulation.

	Panel A: W1 Scores							
	Total Score		Chinese Score		Math Score		English Score	
	Main	Interaction	Main	Interaction	Main	Interaction	Main	Interaction
Peer High Parental Aspiration	0.972** (0.463)	-0.095 (0.537)	0.762** (0.347)	0.074 (0.397)	0.338 (0.459)	-0.778 (0.565)	0.466 (0.436)	-0.743 (0.595)
Parental High Aspiration	0.452*** (0.060)	-0.517 (0.314)	0.353*** (0.054)	-0.272 (0.253)	0.401*** (0.059)	-0.666** (0.303)	0.433*** (0.061)	-0.641 (0.407)
Student Aspiration	0.461*** (0.042)	0.465*** (0.041)	0.404*** (0.043)	0.406*** (0.044)	0.387*** (0.050)	0.388*** (0.049)	0.399*** (0.049)	0.403*** (0.048)
Peer × Parental High Aspiration	– –	1.294*** (0.414)	– –	0.840** (0.344)	– –	1.413*** (0.401)	– –	1.407** (0.532)
	Panel B : W2 Scores							
	Total Score		Chinese Score		Math Score		English Score	
	Main	Interaction	Main	Interaction	Main	Interaction	Main	Interaction
Peer High Parental Aspiration	1.145** (0.521)	0.278 (0.638)	1.551*** (0.396)	0.889** (0.422)	0.629 (0.591)	-0.457 (0.718)	0.719 (0.524)	-0.198 (0.642)
Parental High Aspiration	0.450*** (0.062)	-0.336 (0.308)	0.351*** (0.060)	-0.250 (0.282)	0.414*** (0.068)	-0.625** (0.302)	0.423*** (0.060)	-0.392 (0.328)
Student Aspiration	0.479*** (0.043)	0.482*** (0.043)	0.419*** (0.046)	0.421*** (0.046)	0.415*** (0.047)	0.416*** (0.046)	0.423*** (0.048)	0.426*** (0.049)
Peer × Parental High Aspiration	– –	1.050** (0.425)	– –	0.809** (0.374)	– –	1.375*** (0.418)	– –	1.067** (0.449)
Observations	3003		2865		2940		2860	

Note: This table examines whether peer parental aspirations exert differential effects on academic performance across subject-specific outcomes while controlling subject-specific teacher characteristics. The outcome variable in the first 2 columns are the total standardized academic score, as assessed by the homeroom teacher control. The remaining columns use subject-specific standardized scores controlled by respective subject teacher controls. Cols 1-2 uses home teacher controls, while Cols 3 to 8 substitute controls for Chinese, Math, and English teacher controls, respectively. All specifications retain the full set of student, peer, and

teacher controls as well as SchGrade fixed effects. Robust standard errors clustered at the school-grade level are shown in parentheses; *: $p < 0.1$, **: $p < 0.05$, ***: $p < 0.01$.

Table 6: Effects of Peer Parental and Parental Aspirations by Subject

7.3 Gender

	PanelA: W1Scores			
	Girls		Boys	
	Main	Interaction	Main	Interaction
Peer High Parental Aspiration	0.738 (0.523)	-0.319 (0.693)	1.481** (0.699)	0.374 (0.849)
Parental High Aspiration	0.535*** (0.079)	-0.413 (0.343)	0.378*** (0.087)	-0.612 (0.460)
Student Aspiration	0.435*** (0.069)	0.434*** (0.067)	0.469*** (0.069)	0.476*** (0.069)
Peer × Parental High Aspiration	–	1.250** (0.484)	–	1.332** (0.598)
	Panel B : W2 Scores			
	Girls		Boys	
	Main	Interaction	Main	Interaction
Peer High Parental Aspiration	0.708 (0.477)	-0.429 (0.677)	1.921** (0.785)	1.314 (0.991)
Parental High Aspiration	0.478*** (0.083)	-0.542 (0.388)	0.438*** (0.078)	-0.105 (0.433)
Student Aspiration	0.464*** (0.069)	0.464*** (0.067)	0.478*** (0.071)	0.482*** (0.071)
Peer × Parental High Aspiration	–	1.345** (0.543)	–	0.731 (0.593)
Observations	1504		1499	

Note: This table presents gender-differentiated effects of peer and parental aspirations on academic performance. The “Main” columns show results from additive specifications, while “Interaction” columns include peer-parental aspiration interaction terms. All specifications include the full set of student characteristics (age, minority status, hukou, migration status, siblings, kindergarten attendance, primary school progression), peer characteristics (gender composition, migration status, repetition rate, persistence, and noncognitive skills), and

teacher characteristics (age, gender, marital status, education, experience, and certification), as well as SchGrade fixed effects. Robust standard errors clustered at the school-grade level are shown in parentheses; *: $p < 0.1$, **: $p < 0.05$, ***: $p < 0.01$.

Table 7: Effects of Peer and Parental Aspirations by Gender and Specification

Table 7 reveals gender heterogeneity in the aspiration effect. Boys benefit strongly and consistently from PPA in additive models, though in interaction models their sensitivity to aspiration alignment diminishes over time. This suggests boys' academic outcomes are primarily driven by broad exposure to high-aspiration peer environments, regardless of family context. However, girls show no significant standalone peer effects but demonstrate robust sensitivity to the interplay between peer and family environments, with strong positive interactions when both aspirations align and negative effects when peer aspirations outpace family support. This pattern indicates girls' performance depends critically on coordinated reinforcement across social spheres rather than peer exposure alone.

These findings challenge uniform policy approaches, suggesting boys benefit universally from high-PPA environments while girls require matched peer family aspiration alignment to thrive academically. The fact that boys' peer effects strengthen while the interaction effects fade suggests peer influences operate through different mechanisms by gender: environmental exposure for boys versus targeted reinforcement of family expectations for girls. Methodologically, the contrast between additive and interactive models underscores how omitting interaction terms can mask critical heterogeneity, highlighting the importance of modeling gender-specific processes in educational attainment research.

8 Robustness

8.1 Subsample with limited exposure

To address concerns that peer environment might influence student or parental responses, I restrict the analysis to a subsample of students surveyed immediately following classroom enrollment. At this early stage, students and parents would have had minimal exposure to their new peer group, thereby limiting the possibility of reverse causality or peer-induced bias in reported aspirations or attitudes.

As shown in Table 1, the 1,623 students surveyed in Fall 2013 constitute a valid random sample, as their characteristics (including school location, type, and family wealth) closely mirror those of the overall sample of 3,003 students. This comparability, together with the random assignment to classrooms, helps ensure that the findings are generalizable.³⁰ The phased data collection driven by logistical factors rather than systematic regional variations, ensures the multi-stage probability proportional to size (PPS) sampling design remains nationally representative.

	PanelA:W1Scores			
	Full sample		2013 Fall	
	Main	Interaction	Main	Interaction
Peer High Parental Aspiration	0.972** (0.463)	-0.095 (0.537)	0.264 (0.669)	-1.072 (0.716)
Parental High Aspiration	0.452*** (0.060)	-0.517 (0.314)	0.445*** (0.078)	-0.771** (0.334)
Student Aspiration	0.461*** (0.042)	0.465*** (0.041)	0.422*** (0.057)	0.426*** (0.055)
Peer × Parental High Aspiration	– –	1.294*** (0.414)	– –	1.600*** (0.435)

³⁰ Although the 2014 Spring sample includes a higher proportion of private schools relative to the 2013 Fall sample, the fixed effects approach effectively accounts for these differences.

	Panel B : W2 Scores			
	Full sample		2013Fall	
	Main	Interaction	Main	Interaction
Peer High Parental Aspiration	1.145** (0.521)	0.278 (0.638)	0.682 (0.657)	-0.550 (0.804)
Parental High Aspiration	0.450*** (0.062)	-0.336 (0.308)	0.422*** (0.085)	-0.700* (0.408)
Student Aspiration	0.479*** (0.043)	0.482*** (0.043)	0.455*** (0.055)	0.459*** (0.054)
Peer × Parental High Aspiration	– –	1.050** (0.425)	– –	1.476** (0.558)
Observations	3003		1623	

Note: This table presents estimates based on a restricted sample of students surveyed immediately after Fall 2013 classroom enrollment. Baseline estimates from the full sample (Columns 1-2) are included for comparison. Formal Chow tests examining coefficient equality across samples yield F-statistics of 0.84 ($p=0.364$) for the Wave 1 interaction term and 0.59 ($p=0.445$) for Wave 2, indicating we cannot reject stability of complementarity effects. Joint tests of all aspiration coefficients similarly show no significant structural differences (Wave 1: $F=1.30$, $p=0.285$; Wave 2: $F=0.34$, $p=0.796$). All models control for student, peer and teacher controls, and include SchGrade fixed effects. Robust standard errors clustered at the school-grade level are shown in parentheses; *: $p < 0.1$, **: $p < 0.05$, ***: $p < 0.01$.

Table 8: Robustness Check: Subsample with Limited Exposure

Table 8 presents a robustness check with this subsample. The effect of PPA with subsample failed to achieve statistical significance through formal equivalence tests, it may reflect substantive differences in effect dynamics. Point estimates exhibited a notable downward trend alongside reduced precision (W1 from 0.972** to 0.264; W2 from 1.145** to 0.682), consistent with two potential interpretations. Firstly, the pervasive peer environment effect may require sustained classroom interaction to fully manifest, whereas aspiration complementarity operating through immediate social comparison and identification mechanisms emerges rapidly upon class formation. Secondly, smaller sample sizes reduce statistical power for detecting main effects of moderation, whereas larger and more precise interaction terms remain detectable.

The PPA effects in Interaction Models are more negative in the subsample, suggesting that in early classroom formation, students from low-aspiration

families may initially struggle in high-aspiration environments. Moreover, the parental aspiration coefficient is more strongly negative, which when combined with the larger interaction terms, implies an even steeper gradient of returns to peer academic performance based on family background in the initial classroom period. This suggests complementary benefits between peer and family inputs require time to materialize fully. With stable student aspiration coefficients across specifications, it validates these patterns.

However, the interaction effect demonstrated robustness across both study phases and different exposure durations (W1 1.600*** vs 1.294***; W2 1.476** vs 1.050**, all p-values exceeding 0.36 in Chow tests), indicating that the benefits of aspiration alignment between peer and family contexts emerge rapidly during early class formation and persist.³¹ Interaction models reveal nuanced dynamics: high parental aspiration exerts negative pressure in low-aspiration peer environments, particularly during initial classroom formation.

8.2 Other Checks

Three additional robust checks are provided in the Appendix. Firstly, to ensure that the observed PPA effect did not merely reflect differences in peer parents' educational attainment and family wealth, Table A3 incorporated peer parents' educational level and family wealth into the model. Notably, PPA coefficient retained statistical significance and increased slightly in magnitude with educational background control alone and remained significantly positive with both educational background control and family wealth control. This pattern suggests that the PPA effect reveals a distinct motivational or normative mechanism, rather than stemming from the socioeconomic advantages implied by peer parents' educational attainment and family wealth.

³¹ Formal Chow tests strongly support coefficient stability across samples: Cannot reject equality of the interaction coefficient in either Wave 1 ($F=0.84$, $p=0.364$) or Wave 2 ($F=0.59$, $p=0.445$). Joint tests of all aspiration-related coefficients similarly reveal no statistically significant structural differences between samples (Wave 1: $F=1.30$, $p=0.285$; Wave 2: $F=0.34$, $p=0.796$).

To address concerns regarding parental interference in classroom assignments, Table A4 restricts the analysis to schools where principals explicitly confirm no parental influence over class placements. The results demonstrate remarkable robustness. This consistency across samples with verified randomization confirms that the complementary effects of peer parental and family aspirations represent structural relationships. It reinforces that benefits from high PPA concentrate among students with ambitious parents.

Second, I test the robustness of the results to potential influences from extreme classroom sizes, following a similar method as Yu (2020). Table A5 restricts the sample to classrooms with 21-60 students, eliminating potential distortions from extreme class sizes where peer interactions or teacher attention allocation might confound results. The key interaction effects remain strongly significant with magnitudes slightly exceeding baseline estimates, confirming that family-society complementarity reflects fundamental social dynamics rather than classroom management artifacts. Within this moderate size range, peer effects for low-aspiration students become relatively more beneficial, suggesting extreme class sizes may exacerbate disadvantages for vulnerable populations.

9 Mechanism

As shown in Figure 5, PPA may work through many channels: Direct student influence; Indirect peer, parents, and teachers' influence.

9.1 Direct Effect on Students

Students' Own Educational Aspirations: When examining Wave 2 students' own educational aspirations (from 1 ("Drop out now") to 9 ("Get a Doctor degree")), the analysis reveals a notable and statistically significant positive impact from peer parental aspirations.³² As shown in Figure 6 and Table A6 Panel A, a positive shift in peer parental aspiration is associated with an increase in students' self-reported educational expectations. Together with recent evidence from Cai et al. (2025) on classroom inequality effects. Our findings on aspiration internalization align with the social learning mechanisms described by Bandura (1977).

³² In the survey, a small number (99 out of 3003) of respondents (coded as 10) indicated "don't care" when asked about their expected highest level of education. These responses were recoded to the median category (5 = "Go to senior high school") to retain them in the analysis without disproportionately influencing the distribution. This approach assumes that a non-response of this kind reflects uncertainty rather than systematically higher or lower aspirations.

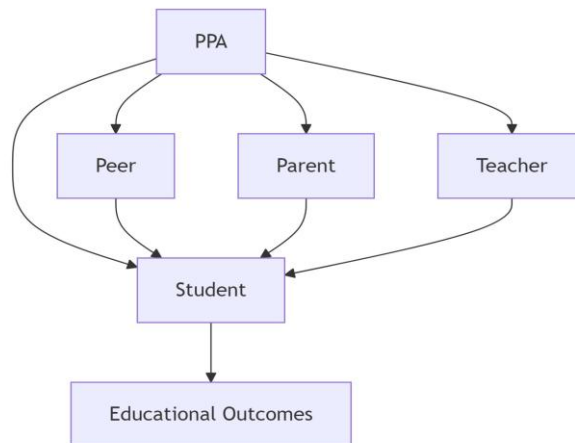


Figure 5: Multichannel Mechanism of PPA

Perceptions of Parental Support and Confidence: Students were asked to evaluate the extent of parental support with the question “How much do you think your parents have done for you?” with response options ranging from 1 (“Very little”) to 5 (“A great deal”).³² In Wave 2, there was a question: “Are your parents confident about your future?” with response options ranging from 1 (“Not confident at all”) to 4 (“Very confident”). As shown in Figure 6 and Table A6 Panel A, peer parental aspirations exert a significant influence on students’ perceptions of parental support and confidence, whereas the direct effect of their own parents’ aspirations appears statistically negligible in these domains. This pattern, consistent across both perceptual outcomes, indicates that peer influences operate independently of familial inputs rather than interacting with them.

These results demonstrate that educational aspirations and perceptions of parental support are socially constructed within peer networks, not just shaped within the family. Two key mechanisms appear: direct modeling of peer group norms and social comparison against a peer benchmark. This influence is most

³²Wave 1 responses are used here due to data availability, there were no such questions in W2.

potent for students lacking strong parental support, suggesting a compensatory role for peer context. Therefore, understanding how parental aspirations affect student outcomes requires integrating the powerful influence of the social environment.

9.2 Indirect Effect Through Peer

When peer controls are introduced in the Baseline models, the magnitude of the effect of peer parental aspirations on student performance declines substantially (as in Table 4). This pattern indicates that PPA operate through peer-related channels. Specifically, it could cultivate a more supportive and academically oriented culture, enhancing students' sense of belonging.

This interpretation is supported by Figure 6 and Table A6 Panel B. Students in high-PPA classrooms report a more positive environment, consistent with the findings of Craig and Martin (2023) that improved school culture is linked with better scores. They experience a stronger sense of belonging and classroom cohesion, feel closer to their school community, and participate more in school activities. Furthermore, their peer networks are characterized by stronger academic performance, greater study effort, and higher university expectations. Critically, the absence of significant interactions suggests these beneficial effects operate broadly, rather than being concentrated among specific subgroups. Taken together, PPA fosters a more positive and academic-oriented peer environment, which may serve as a key transmission channel linking peer backgrounds to academic outcomes.

9.3 Indirect Effect Through Parent

PPA may shape student achievement by altering parental behavior and perceptions. As shown in prior literature, parents often adjust their investment based on perceptions of their child's academic context (i.e. Bergman, 2021; Abdulkadiroglu et al., 2014; Kinsler and Pavan, 2021). In line with this, Table A6 Panel C shows that PPA increases parental engagement in academic supervision, particularly in

homework-related activities. However, this effect is moderated by family aspiration: negative interaction terms suggest that the influence of PPA is weaker among students with high parental aspirations, pointing to a substitution pattern.

Bergman (2021) and Behrman et al. (2024) have demonstrated parental investment behaviors are responsive to children's performance. However, there is limited evidence that high PPA increases financial investment in education, such as shadow education attendance or cost. Instead, parental adjustments appear concentrated in non-material domains, most notably through increased involvement in academic oversight and greater confidence in their child's future, as shown in Figure 6 and Table A6 Panel C. The negative interaction observed in aspiration related outcomes further indicates that peer influences primarily benefit students whose parents initially had lower educational aspirations. Together, these results suggest that PPA acts as a compensatory mechanism, encouraging low-aspiration families to be more engaged and supportive, particularly in domains such as homework monitoring and future confidence rather than monetary means.

9.4 Indirect Effect Through Teacher

Another possible channel is teacher-related perceptions and behaviors. As shown in Figure 6 and Table A6 Panel D, exposure to higher PPA leads parents to perceive teachers as more responsible and marginally more patient. The absence of significant interaction effects indicates this positive shift in perception occurs broadly across different family backgrounds, not just within specific subgroups.

In contrast, student-reported measures of teacher behavior, such as the frequency of praise or criticism, show no significant response to PPA. This clear divergence between parental perceptions and student reports suggests that PPA operates through a social-psychological channel, influencing how parents interpret the educational environment rather than changing actual classroom dynamics. Ultimately, these findings indicate that PPA reinforces positive parental assessment toward teachers, which may subsequently influence parental involvement or advocacy, without directly affecting student-teacher interactions.

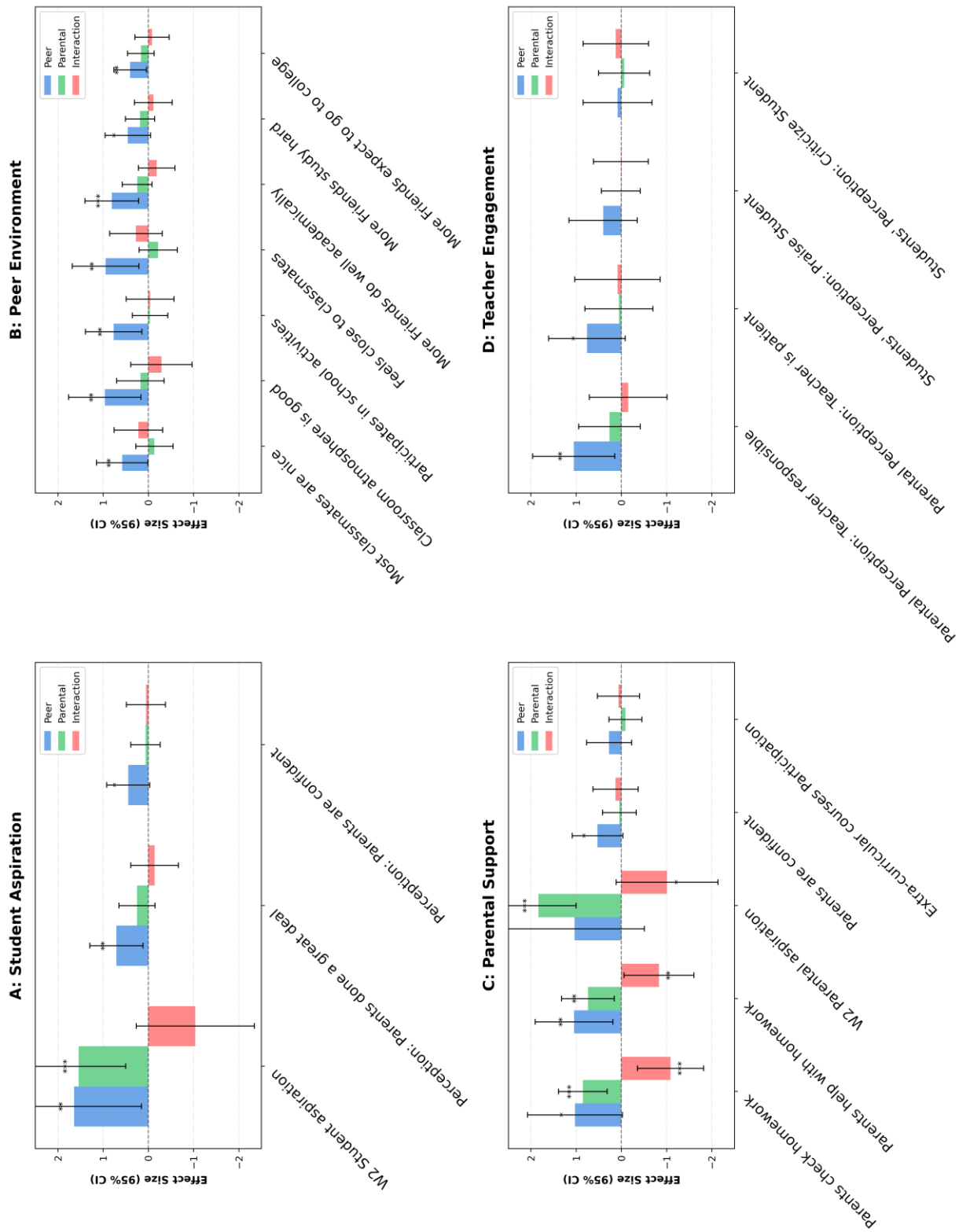


Figure 6: Multichannel Mechanism of PPA

Note: This figure plots the coefficients from separate regressions of the listed outcomes on peer parental aspiration, on parental aspiration, and their interaction, with student aspiration included as a covariate. All models control for student, peer, and teacher characteristics and include school-grade fixed effects. Standard errors are at school-grade level.

10 Conclusion

This study extends models of educational production by demonstrating that parental inputs operate not only within the household but also through classroom peer networks. Leveraging unique microdata with quasi-random assignment, I provide robust evidence that peer parental aspirations are a critical dimension of educational production, generating effects distinct from traditional peer measures.

There are three principal insights. First, peer parental aspiration constitutes a direct and robust input into educational production, improving student achievement. Second, these effects operate through multiple pathways, including student norm internalization, classroom environment, parental investment, and exhibit strong complementarity with own family context, disproportionately benefiting those with high-aspiration parents. Third, the results challenge the notion of uniform peer effects, demonstrating that returns are heterogeneous and contingent on family inputs, revealing how cumulative advantages in high aspiration environments primarily benefit already-advantaged students.

These findings carry important policy implications for educational inequality. The persistent lack of peer benefits for disadvantaged students aligns with Ding and Lehrer (2007)'s finding that high-ability students gain more from better peers, limiting the potential of peer-mixing to reduce achievement gaps. This could be a reason why better baseline peers often benefit more from interventions. These patterns highlight the enduring role of family-society complementarities, suggesting effective policy should target aspiration gaps through information dissemination (e.g., about university returns as in Bleemer and Zafar (2018)) or programs leveraging high-aspiration networks as catalysts for behavioral change.

Future research should explore whether these peer effects persist in long-term outcomes like university attendance and labor market success. More fundamentally, the results call for re-conceptualizing educational production to account for the inter-generational transmission of aspirations, not just individual family investments. By mapping these complex transmission channels, this study advances our understanding of how unpacking peer effects may help stop inequality reproduction across generations.

11 Appendix

11.1 Main Results

11.1.1 Missing observation

	Peer Parental Aspiration Levels	
	Non-missing	Missing
Mean	0.893	0.882
Standard Error	0.003	0.005
Observations	3.003	1.322
Difference	0.011	
p-value	0.051	

Note: This table compares peer parental aspiration levels between students with complete data (non-missing) and those with missing data. The difference of 0.011 units approaches statistical significance ($p = 0.051$), suggesting a borderline systematic pattern where students with missing data have slightly lower peer aspiration exposure. The 95% confidence interval indicates the true difference could range from negligible to substantively meaningful levels.

Table A1: T-Test of Peer Parental Aspiration Levels by Missing Data Status

The t-test in Table A1 reveals a borderline statistically significant difference in peer parental aspiration levels between students with complete and missing data. Students with non-missing data show slightly higher peer aspiration levels (0.893) compared to those with missing data (0.882), with a difference of approximately 0.011 units on the 0-1 scale. The p-value of 0.051 indicates this difference approaches conventional significance thresholds, suggesting a potential systematic pattern where students with lower peer aspiration exposure may be somewhat more likely to have missing data. While this suggests missingness is not completely random, the small effect size and borderline significance indicate that any resulting bias in peer effect estimates is likely limited.

11.1.2 Main model without Student Aspiration

	W1		W2	
	Main	Interaction	Main	Interaction
Peer High Parental Aspiration	1.058** (0.465)	0.053 (0.568)	1.234** (0.522)	0.432 (0.669)
Parental High Aspiration	0.653*** (0.066)	-0.258 (0.346)	0.659*** (0.066)	-0.068 (0.338)
Peer × Parental High Aspiration	– –	1.219*** (0.450)	– –	0.973** (0.457)
Observations	3003	3003	3003	3003

Note: This table presents effects of peer and parental aspirations on academic performance without “Student Aspiration” variable. All specifications include the full set of student characteristics (age, minority status, hukou, migration status, siblings, kindergarten attendance, primary school progression), peer characteristics (gender composition, migration status, repetition rate, persistence, and noncognitive skills), and teacher characteristics (age, gender, marital status, education, experience, and certification), as well as SchGrade fixed effects. Robust standard errors clustered at the school-grade level are shown in parentheses; *, $p < 0.1$, **, $p < 0.05$, ***, $p < 0.01$.

Table A2: Baseline model and interaction model without Student Aspiration

Table A2 reports the effects of peer parental aspirations and parental aspirations on academic performance in specifications that exclude the Student Aspiration variable. The estimated coefficients are broadly consistent with those obtained from models that have student aspirations, indicating that the influence of peer parental aspirations on academic achievement is robust to this exclusion.

11.1.3 Variation of PA

The histogram Figure A1 reveals substantial variation in classroom Parental Aspiration. It displays the distribution of Parental Aspiration (PA) across classrooms, showing the percentage of classrooms (y-axis) containing varying proportions of high-aspiration parents (x-axis).

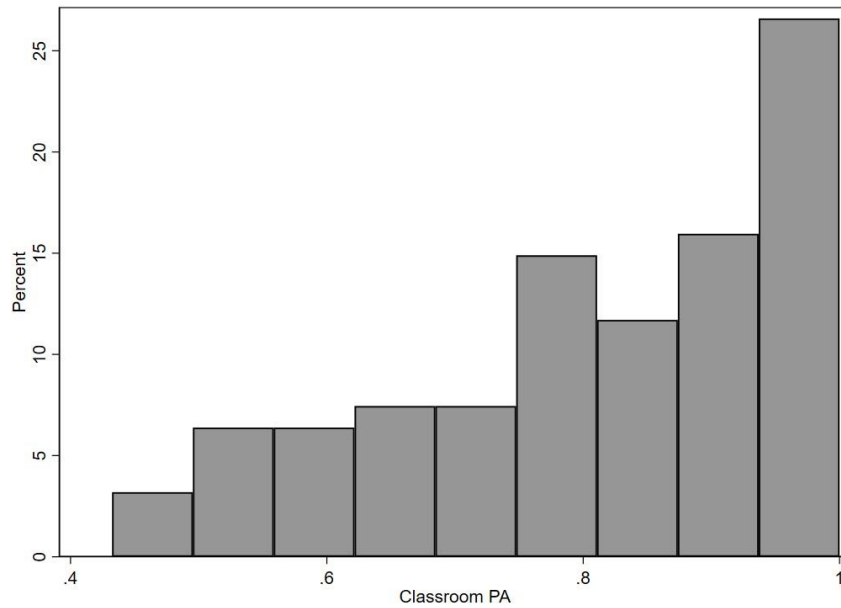


Figure A1: Histogram for Class Parental Aspiration

11.2 Robustness checks

11.2.1 Peer Parental Education and Wealth

Table A3 provides clear evidence that peer parental aspirations exert a robust and independent influence on student achievement, distinct from the effect of peer parental education and family wealth. In both W1 and W2, the estimated coefficients for peer high parental aspiration remain positive and statistically significant, even after controlling for the average educational attainment of peers' parents and peer family wealth. The results suggest that what matters for student performance is not the educational background of peer families or family wealth, but rather the normative aspirations and expectations they convey. This distinction reinforces the interpretation of aspirations as a motivational signal transmitted through social networks rather than a mere proxy for socioeconomic status. The consistency of results across waves further indicates that the influence of aspirational peer environments is persistent and structurally embedded in classroom interactions, validating the study's causal claims.

Panel A: Wave 1 Scores				
	Baseline	With Peer Parental Edu	With Peer Family Wealth	With Both
Peer High Parental Aspiration	0.972** (0.463)	1.027** (0.463)	0.947* (0.477)	0.987** (0.476)
Parental High Aspiration	0.452*** (0.060)	0.454*** (0.060)	0.451*** (0.061)	0.452*** (0.061)
Student Aspiration	0.461*** (0.042)	0.460*** (0.042)	0.461*** (0.042)	0.460*** (0.042)
Peer Parental Education	– –	-0.336 (0.365)	– –	-0.382 (0.374)
Peer Family Wealth	– –	– –	0.090 (0.332)	0.163 (0.344)
Panel B: Wave 2 Scores				
	Baseline	With Peer Parental Edu	With Peer Family Wealth	With Both
Peer High Parental Aspiration	1.145** (0.521)	1.150** (0.535)	1.102* (0.564)	1.110* (0.571)
Parental High Aspiration	0.450*** (0.062)	0.450*** (0.062)	0.448*** (0.062)	0.449*** (0.063)
Student Aspiration	0.479*** (0.043)	0.479*** (0.043)	0.479*** (0.043)	0.479*** (0.043)
Peer Parental Education	– –	-0.033 (0.524)	– –	-0.079 (0.521)
Peer Family Wealth	– –	– –	0.149 (0.446)	0.164 (0.455)
All controls	Y	Y	Y	Y
Observations	3003	3003	3003	3003

Note: This table demonstrates peer parental aspiration's robust influence across waves regardless of parents' own education. Peer Parental Education is leave-me out average of a binary indicator (1 = at least one parent holds a university degree or higher, 0 = otherwise). Panel A shows Wave 1 results where increase in peer aspiration raises scores unaffected by adding parental education. Panel B replicates this independence in Wave 2. All models include school-grade fixed effects and full controls. Robust standard errors clustered at the school-grade level are shown in parentheses; *: $p < 0.1$, **: $p < 0.05$, ***: $p < 0.01$.

Table A3: Peer Parental Aspiration Effects and Peer Parental Education

11.2.2 Parental Interference

Table A4 addresses potential concerns about parental interference in classroom assignments by restricting the sample to schools where principals confirm no parental influence over class placements. The results remain robust, with the interaction between peer and parental aspirations maintaining strong statistical significance (1.492*** for Wave 1 and 1.234** for Wave 2), even slightly larger in magnitude than the full sample estimates (1.294*** and 1.050, respectively). This consistency confirms that the core findings are not driven by non-random sorting, as the complementary effect of peer and family aspirations persists in settings where randomization is strictly enforced. The stability of student aspiration coefficients

Panel A : W1 Scores		
	Full sample	No Parental interference subsample
Peer High Parental Aspiration	-0.095 (0.537)	-0.308 (0.579)
Parental High Aspiration	-0.517 (0.314)	-0.642* (0.346)
Student Aspiration	0.465*** (0.041)	0.479*** (0.045)
Peer High Parental Aspiration × Parental High Aspiration	1.294*** (0.414)	1.492*** (0.458)
Panel B : W2 Scores		
	Full sample	No Parental interference subsample
Peer High Parental Aspiration	0.278 (0.638)	0.144 (0.686)
Parental High Aspiration	-0.336 (0.308)	-0.425 (0.329)
Student Aspiration	0.482*** (0.043)	0.491*** (0.050)

Peer High Parental Aspiration × Parental High Aspiration	1.050** (0.425)	1.234** (0.464)
School-grade fixed effect	Y	Y
Student controls	Y	Y
Peer controls	Y	Y
Teacher controls	Y	Y
Observations	3003	2540

Note: This table presents estimates based on a restricted sample that excludes schools where parents may have influenced classroom assignment decisions, addressing concerns about non-random sorting. Baseline coefficients from the full sample are included in Col 1 for comparison. All specifications retain the full set of student, peer, and teacher controls as well as SchGrade fixed effects. Robust standard errors clustered at the school-grade level are shown in parentheses; *: $p < 0.1$, **: $p < 0.05$, ***: $p < 0.01$.

Table A4: Robustness Check: Subsample Excluding Schools with Parental Influence on Classroom

(0.479* and 0.491***) further supports the interpretation that family-society synergies are structural rather than artifacts of selection bias.

The subsample analysis reinforces the study’s causal claims: even after excluding schools with potential parental interference, the peer-parental interaction effect remains economically and statistically significant. The similarity in coefficients across samples suggests that the main results generalize to contexts where randomization is rigorously implemented. Notably, the negative main peer effects for low-aspiration students persist (-0.308 in W1, 0.144 in W2), though they remain statistically insignificant, highlighting that the benefits of high-aspiration peers are concentrated among students with similarly ambitious parents. These results underscore that the identified peer effects reflect genuine social spillovers rather than confounding from parental manipulation of classroom composition.

11.2.3 Class Size

To systematically address concerns regarding extreme classroom sizes potentially affecting the results, I examine the distribution of class sizes through histogram analysis and implement targeted sample restrictions.

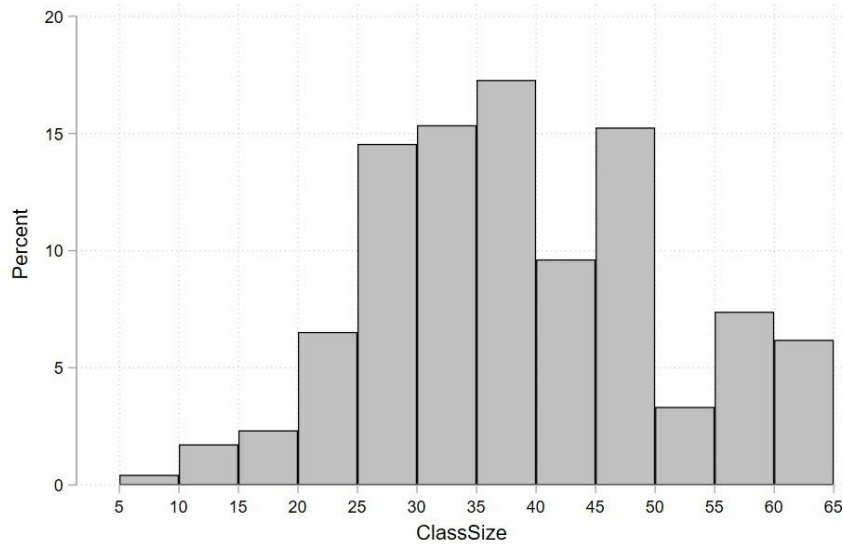


Figure A2: Histogram for Class Size

The histogram Figure A2 reveals substantial variation in classroom enrollment, with some classes falling outside typical operational ranges that might confound peer effect estimates through altered classroom dynamics or teacher attention allocation. Consequently, I restrict the analytical sample to classrooms with enrollments between 21 and 60 students, eliminating both unusually small classes where peer interactions may be limited and exceptionally large classes where classroom management challenges could obscure genuine peer effects.³³

³³ Small class is less common in China, there are some schools, especially in more developed areas, that may have classes fewer than 25 students. A significant percentage of classes, especially in some province, have more than 55 students (see <https://www.sixthtone.com/news/1000618>).

	Panel A : W1 Scores Class	
	Full sample	Size 21-60
Peer High Parental Aspiration	-0.095 (0.537)	-0.053 (0.658)
Parental High Aspiration	-0.517 (0.314)	-0.581* (0.324)
Student Aspiration	0.465*** (0.041)	0.480*** (0.043)
Peer High Parental Aspiration × Parental High Aspiration	1.294*** (0.414)	1.408*** (0.428)
	Panel B : W2 Scores	
	Full sample	Class Size 21-60
Peer High Parental Aspiration	0.278 (0.638)	0.501 (0.764)
Parental High Aspiration	-0.336 (0.308)	-0.342 (0.324)
Student Aspiration	0.482*** (0.043)	0.493*** (0.049)
Peer High Parental Aspiration × Parental High Aspiration	1.050** (0.425)	1.082** (0.451)
School-grade fixed effect	Y	Y
Student controls	Y	Y
Peer controls	Y	Y
Teacher controls	Y	Y
Observations	3003	2682

Note: This table presents regression estimates based on a subsample that excludes classrooms with extremely small or large sizes to reduce potential bias from atypical peer environments. Col 1 reports the baseline estimates from the full sample for comparison. All specifications include school-grade fixed effects and control for student, peer, and teacher characteristics. Robust standard errors clustered at the school-grade level are shown in parentheses; *: $p < 0.1$, **: $p < 0.05$, ***: $p < 0.01$.

Table A5: Robustness Check: Subsample Excluding Classrooms with Extremely Small or Large Sizes

Table A5 addresses concerns about extreme classroom sizes by restricting the sample to classes with 21-60 students, eliminating potential distortions from unusually small or large class environments. The results show remarkable consistency with the full sample, as the key interaction between peer and parental aspirations remains strongly significant (1.408*** for Wave 1 and 1.082** for Wave 2) and even increases slightly in magnitude compared to baseline estimates. This stability confirms that the core finding of complementary family-society effects is not driven by atypical class sizes where peer interactions might be diluted or teacher attention disproportionately allocated. The persistence of these patterns suggests the estimated peer effects reflect fundamental social dynamics rather than artifacts of classroom management challenges.

The robustness of results across class size specifications reinforces the study's causal claims while offering practical insights for education policy. The slightly larger interaction effects in the restricted sample (particularly for Wave 1) may indicate that family-society complementarity operates most strongly in mid-sized classrooms that balance sufficient peer diversity with manageable group dynamics. Notably, the peer parental effects for low-aspiration students become relatively more beneficial in the restricted sample (-0.053 in W1, 0.501 in W2), suggesting extreme class sizes might exacerbate disadvantages for these students. These findings imply that while peer effects are robust across typical classroom contexts, policymakers should consider optimal class sizes when designing interventions to maximize the benefits of positive peer environments, particularly for students from less-advantaged backgrounds.

12 Mechanism

The results in Table A6 show that higher peer parental aspirations shape students' expectations and achievement mainly through social and relational channels: they elevate students' own educational aspirations and perceptions of parental support, foster more positive peer norms and classroom climate, and induce additional nonmonetary parental investments such as homework supervision and confidence

in the child's future, while leaving teacher behavior largely unchanged and instead shifting how parents perceive teachers, with these effects operating broadly but especially benefiting students from initially low-aspiration families.

	Peer High Parental Aspiration	Parental High Aspiration	Peer $\times$ Parental High Aspiration
Panel A: Student Aspiration			
W2 Student Aspiration	1.638**	1.537***	-1.039
Perception: Parents have done a great deal	0.704**	0.250	-0.140
Perception: Parents are confident	0.444*	0.060	0.051
Panel B: Peer Environment			
Most classmates are nice	0.575*	-0.137	0.218
Classroom atmosphere is good	0.962**	0.174	-0.292
Participates in school activities	0.765**	-0.040	-0.041
Feels close to classmates	0.945**	-0.218	0.272
More Friends do well academically	0.807**	0.246	-0.186
More Friends study hard	0.453*	0.181	-0.113
More Friends expect to go to university	0.400**	0.162	-0.085
Panel C: Parental Support			
Parents check homework	1.025*	0.853***	-1.090***
Parents help with homework	1.044**	0.738**	-0.830**
W2 Parental aspiration	1.035	1.834***	-1.010*
Parents are confident	0.526*	0.044	0.128
Extra-curricular courses Participation	0.271	-0.091	0.062
Extra-curricular courses Cost	-31.385	631.333	-1020.929
Panel D: Teacher Engagement			
Parental Perception: Teacher responsible	1.051**	0.262	-0.151
Parental Perception: Teacher is patient	0.761*	0.051	0.086
Students' Perception: Praise Student	0.403	0.012	0.010
Students' Perception: Criticize Student	0.082	-0.063	0.122

Note: Each row reports coefficients from a separate regression of the listed outcome on peer parental aspiration, own parental aspiration, and their interaction, with student aspiration included as a covariate. All models control for student, peer, and teacher characteristics and include school-grade fixed effects. Robust standard errors clustered at the school-grade level are shown in parentheses; *, $p < 0.1$, **, $p < 0.05$, ***, $p < 0.01$.

Table A6: Mechanisms Linking Peer Parental and Parental Aspirations to Student Outcomes

13 Other

Table A7 compares the composition of schools implementing random classroom allocation with those not implementing it, revealing broadly similar characteristics across key dimensions. Randomly allocated schools exhibit a slightly higher proportion of pupils from urban and central areas, alongside marginally elevated household wealth levels; non-randomly allocated schools, conversely, are more concentrated in rural regions. Both types of schools were predominantly public

institutions. These patterns indicate that randomly assigned and non-randomly assigned schools are fundamentally comparable in terms of structure and socio-economic composition, with the degree of difference sufficient to support the external validity of findings derived from the randomized subsample.

Characteristic	Categories	Randomized	Non-Randomized
Student ID	Number of students surveyed	3,003	3313
Class ID	Number of classes represented	94	104
School ID	Number of schools represented	47	52
County ID	Number of counties represented	24	26
School Location	Central area of city or county	1,450 (48.3%)	1370 (41.4%)
	Urban fringe and rural-urban fringe	1,076 (35.8%)	439 (13.3%)
	Towns and rural area	477 (15.9%)	1504 (45.4%)
School Type	Public school	2,799 (93.2%)	3219 (97.2%)
	Private school	204 (6.8%)	94 (2.8%)
Family Wealth	Difficult	357 (11.9%)	818 (24.7%)
	Moderate	2,405 (80.1%)	2340 (70.7%)
	Wealthy	239 (8.0%)	152 (4.6%)

Note: This table reports the student and school composition surveyed in two types during Wave 1: Randomized assignment and Non-randomized assignment. Distributions across school location, school type, and family wealth indicate strong comparability between the two types.

Table A7: W1 Student and School composition

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used ChatGPT, Deepseek and Claude in order to refine language and assist editing. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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